

Engaging Novice VR Users to VR for Education: Enhancing User Comfort, Health and Safety in VR and AR

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ABSTRACT

Despite virtual reality's capacity to impact student engagement learning, and transfer to the physical world, adoption into education has been slow. Our team conducted two one-week summer camps in which students engaged with multiple educational XR tools, completed usability surveys, and engaged in focus groups about the tools. The learner experience design of these camps was done with a focus on safely engaging underrepresented middle school girls with XR experiences they could share with teachers and family members in the future. Camp design choices included safety and selection of experiences.

Keywords: Virtual Reality, Design Thinking, UX, Education.

1 INTRODUCTION

Despite virtual reality's capacity to impact student engagement learning, and transfer to the physical world, adoption into school systems has been slow. Challenges to immersive tool integration include accessibility, usability, and training. Our team hosted two camps (one in summer 2022 and one in summer 2021).

The 2021 summer camp was hosted at a local university campus and a local museum of science and history. The students involved in this program learned about and experienced immersive technologies, augmented reality (AR), and virtual reality (VR), and were exposed to many different XR tools for STEM education. Students were taught to critically evaluate AR & VR tools and will apply evaluations to guide their own design of XR tools to help other young women learn about STEM.

Technology integration does not happen in a vacuum; instead, it is very dependent on contextual elements such as user comfort with the tool, access, infrastructure, teacher experience with the tool, and teacher knowledge of the subject matter [1]. Challenges in usability, comfort, mental effort, operation, physical interaction, cost, and degree of user challenge are unclear when implementing immersive technology in the learning environment (Xi et al., 2022). address these challenges, this mixed methods research study was designed to provide insights into the underrepresented middle and high school students' (ages 11-17), mentors', and teachers' engagement with and perceptions of immersive media tools for learning and student engagement.

The learner-centered design approach to this camp was focused on empowering young women to share what they learned at their schools. Data was collected using the System Usability Scale (SUS), reflections, and focus groups. The findings indicated that the students preferred and indicated that they were more likely to

use technologies that were lower cost and less interactive, due to ease of use. This is meaningful, as many innovators are frequently focused on impressing students with novelty, when they may be more effective in the integration of usable tools.

2 DESIGN OF GIRLS SURGE INTO STEM CAMP

Considerations in the design and delivery of the camp were student learning outcomes, context of the learning experiences, increase self-efficacy, transfer of learning to the students' homes and home schools, and safety. When considering context, the design team looked at factors such as cost, comfort, access to the technology, infrastructure requirements, availability, and cost of applications. Our team worked strategically to balance student safety and privacy with learning and research objectives.

2.1 Design Thinking

The camp content was designed around design thinking theory. We began with introduction to the steps in design thinking: empathize, define, ideate, prototype, test. During the five-day camp, we walked the students through each aspect of this process. After discussing the process of testing and the importance of testing, we had the students evaluate each experience that we tried using focus group approach and System Usability Scale (SUS).

2.2 Hardware and Software Selection

The first choice in planning this camp was the type of immersive hardware to include in the camp. We introduced the girls to Google Cardboard, Snapchat Spectacles, and the Oculus Quest (and Quest 2 during year 2). We chose these tools to introduce the concept of levels of immersion [2]. We chose openly available tools so students might share with teachers and families in the future. During the camps, students tried Greenland Melting (YouTube VR), The Anne Franke Experience, Mission ISS, and ImmerseMe (language learning via internet browser, donated by the developer to our camp). Each of these experiences was available for free. Students also played VR games during lunch.

2.3 Safety

Because the first iteration of this camp occurred during the COVID 19 pandemic, in the summer of 2021, the team took additional precautions. These precautions included COVID 19 waivers, proof of a recent COVID 19 test, and sanitation. Students wore masks at the summer camp and practiced social distancing, with students sitting together only if they arrived in the same car. Also, equipment (e.g., headsets and hand controllers) were sanitized between users. During the second iteration of the summer camp, we continued to sanitize equipment regularly and provided optional masks. However, the US and state

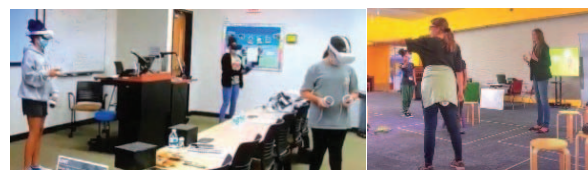


Figure 1: 2021 camp vs 2022 camp.

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of Texas had reduced the COVID 19 public health restrictions, so most students chose not to wear masks or distance.

2.4 UX Research Principles

We applied the theories of experiential learning to UX research. We taught the students about focus groups as qualitative research and surveys as quantitative research. For each learning experience that we used during the camp, we held a focus group and completed the System Usability Scale (SUS).

2.5 Design Thinkers as Speakers

Each day of the camp, students met a woman in STEM who described her journey to her career and gave an example of her application of design thinking professionally or academically. Some of the speakers included a material science engineer, an entomologist, an environmental engineer, a biologist, a computer scientist, a behavioral scientist, and a chemical engineer/entrepreneur. Student engagement was visible and manifested in the numerous questions students asked speakers.

3 OUTCOMES

The summer experience discussed in this paper offers a model for effectively engaging students with emerging technologies in an approach grounded in the design thinking process. Students applied the UX principles throughout and applied the design thinking and understanding of immersive tools and levels of immersion to create their own preliminary designs of immersive interventions to problems that they identified.

3.1 Safety Improvements

In year one and year 2 our team was careful to keep the students separated during VR activities for their safety. During year 2 we used tape to create physical boundaries that mirrored the virtual boundaries for student and teacher safety, e.g. to keep students wearing headsets from accidentally striking a peer or mentor.

3.2 Students Applying UX Principles

It became apparent that the students embraced design thinking on the first day of camp one, in which the students' list of possible apps included discouraging toxic parenting and demonstrating the effects of water pollution. Their subsequent reflection of the importance of testing discussed their perceptions that Chromebook had not been effectively integrated into their classrooms, across school districts.

The principles of comfort and usability resonated with the students throughout both camps, with most focus groups bringing up the level of comfort for the virtual experiences that they tried. Over both years, students were overwhelmingly positive about the comfort and usefulness of the Greenland Melting experience in YouTube VR. They were similarly supportive of future use of The Anne Frank Experience, in which they were guided through a narrated interactive virtual tour of Anne Frank's home. Inversely, students rated the Mission ISS experience poorly in the SUS and in the focus group. While this was the most interactive and immersive experience we used, students reported feeling sick and disoriented. They also complained that the instructions were unclear. This finding suggests that the less immersive experiences (such as the video of the Greenland melting experience) could have more possible applications in the K12 classroom.

3.3 Student Design of Immersive Tools

The final day of the summer camps included students creating a

design of a possible technological intervention for some problem important to them, following the design thinking process. Students also were encouraged to present their design to the class of peers, mentors, and family visitors. All students opted to share their designs and explained their design processes in a public presentation to peers, mentors, and their families.

3.4 Discussion

The camp assessed course and lesson objectives by evaluating students' understanding of the levels of immersion available in XR, application of the SUS to evaluate existing technology, and the application of the design thinking process to identify and address problems they perceive in their daily lives and/or the world. Students reported that they would use what they learned from the camp. Future research will follow up with the students.

4 CONCLUSION

Student feedback was overwhelmingly positive, which was demonstrated in perfect attendance of participants and interactive engagement. Students expressed their gratitude for the opportunity to attend and participate in the camp expressing interest in attending again next year. This experience exposed students to careers in STEM they did not know existed and inspired them to pursue education in the STEM field.

This project is replicable in other communities and contexts. This approach not only provides students access to STEM professionals and emerging technology tools, it also provides an opportunity for research driven iteration on the design, development, implementation, and integration of XR Tools for education and training.

Others could adapt aspects of this project to meet the interest, needs, and experience level of the students. Technological components of this camp, such as the VR headsets, can be replicated with the application for grants and/or use of equipment resources through local universities or edtech companies. The camp could be tailored to include guest speakers that are relevant to the community and goals of the students. The format could be adapted to the community learning spaces available making the day shorter or longer as well as adding or deleting the number of days. Also, resources for the community learning space can be incorporated. For example, we toured the museum and attended presentations led by the museum director. Any program could add learning opportunities that are unique to their community.

ACKNOWLEDGEMENTS

This research was made possible by an American Association of University Women (AAUW) grant, partnership with the Fort Worth Museum of Science History, and support from the University of North Texas College of Information. Special thanks to SURGE XR lab student volunteers who helped with data collection and recruitment (Nelumini Samaranayake, Janna Duke, and Anila Das) and the amazing professionals who spoke at the camp about their experiences as women in STEM.

References

- [1] Niess, M. L. (2011). Investigating TPACK: Knowledge Growth in Teaching with Technology. *Journal of Educational Computing Research*, 44(3), 299–317.
- [2] Hayes, A., Dhimolea, Meng, N., and Tesh, G. (2021). Levels of immersion for language learning from 2D to highly immersive interactive VR.
- [3] Brooke, J. (1996). SUS-A quick and dirty usability scale. *Usability evaluation in industry*, 189(194), 4-7.